



Project report

Galym SEL Fitness

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TASK DISTRIBUTION

SECTION 1		
TASK	Student 1	Student 2
ER	Sahar Hawsawi	Elham Alzahrani
SCHEMA	Safaa Ahmed Sobh	Layan Alshreef
SECTION 2		
TASK	Student 1	Student 2
TRAINEE TABLE AND VALUES	Safaa Ahmed Sobh	-----
STAFF TABLE AND VALUES	Elham Alzahrani	-----
MEMBERSHIP TABLE AND VALUE	Sahar Hawsawi	-----
CONSTRAINTS	Layan Alshreef	-----
SECTION 3		
TASK	Student 1	Student 2
TWO BASIC SELECT	Elham Alzahrani	-----
TWO AGGREGATE FUNCTIONS	Layan Alshreef	-----
ONE JOINS BETWEEN TABLES	Safaa Ahmed Sobh	-----
ONE SUBQUERY, ONE VIEW	Sahar Hawsawi	-----



Section 1



1.1 Project overview

The project focuses on a fitness center management system with four primary entities:

Entities

- **Staff:** Represents employees who manage memberships, perform equipment checks, and interact with trainees. Key attributes include Staff ID, Name (Fname/Lname), Mail, Salary, and Phone.
- **Trainee:** Represents gym members/customers. Key attributes include Trainee ID, Name, Physical Metrics (Height/Weight), Date of Birth, and Phone.
- **Membership:** Represents contractual plans (e.g., Monthly, Yearly, VIP) with specific durations and validity dates.

System Relationships

The system logic is defined by the following key interactions:

- **Trainee & Membership:** A trainees can subscribe to a single membership plan (1:1).
- **Staff & Membership:** A Staff member can be responsible for managing many membership administrations (1:N).
- **Staff & Trainee:** many Trainees can interact with many staff (N:N).



1.2 Entities and Attributes

Staff

Represents the employees of the fitness center.

- Staff_ID (Primary Key - underlined): Unique identifier for each staff member.
- Mail: Email address.
- Salary: The pay rate of the staff.
- Phone: Contact number.
- Name (Composite Attribute): Broken down into Fname (First Name) and Lname (Last Name).

Trainee

Represents the gym members or customers.

- Trainee_ID (Primary Key - underlined): Unique identifier for each member.
- Height & Weight: Physical metrics for tracking progress.
- Date_of_birth: Used for age verification or demographics.
- Phone: Contact number.
- Name (Composite Attribute): Broken down into Fname and Lname.

Membership

Represents the contractual plans members purchase.

- Membership_ID (Primary Key - underlined): Unique identifier for the plan.
- Type: e.g., Monthly, Yearly, VIP.
- Duration: The length of the membership.
- Date (Composite/Derived): Likely the timeframe of the membership, broken into Start_date and Expiration_date.



1.3 Relationships

The diagram uses **N** (many) and **1** (one) notations to define the cardinality.

Relationship	Entities Involved	Description	Cardinality
Manage	Staff & Membership	One Staff member can handle many membership application.	1:N (one-to-Many)
Interacts	Staff & Trainee	Many staff can help a trainee, and a trainee can be helped by many staff.	N:N (Many-to-Many)
Subscribes	Trainee & Membership	A trainee can have one membership at a time.	1:1 (1 Trainee to 1 Membership)



1.4 ER Diagram

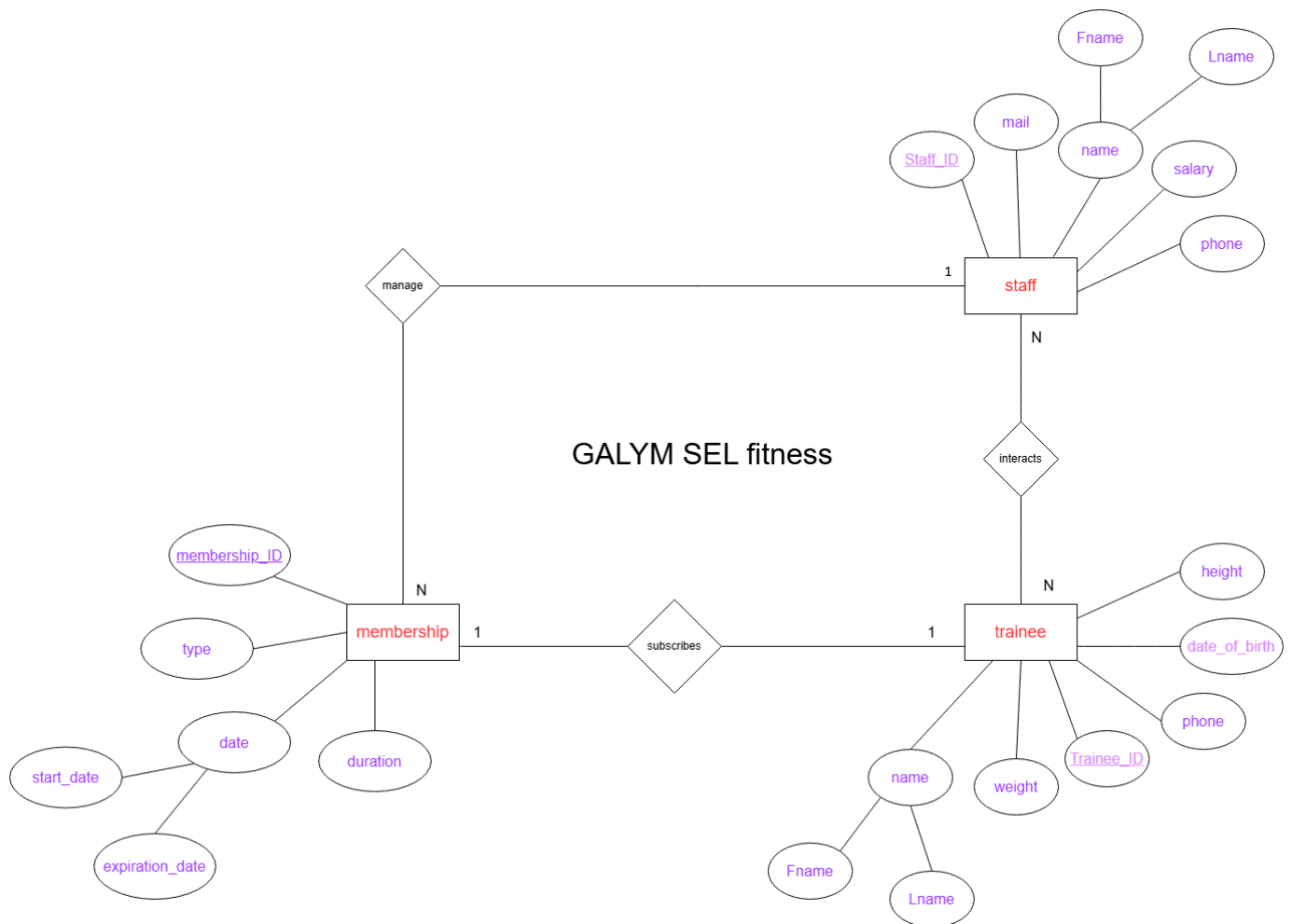
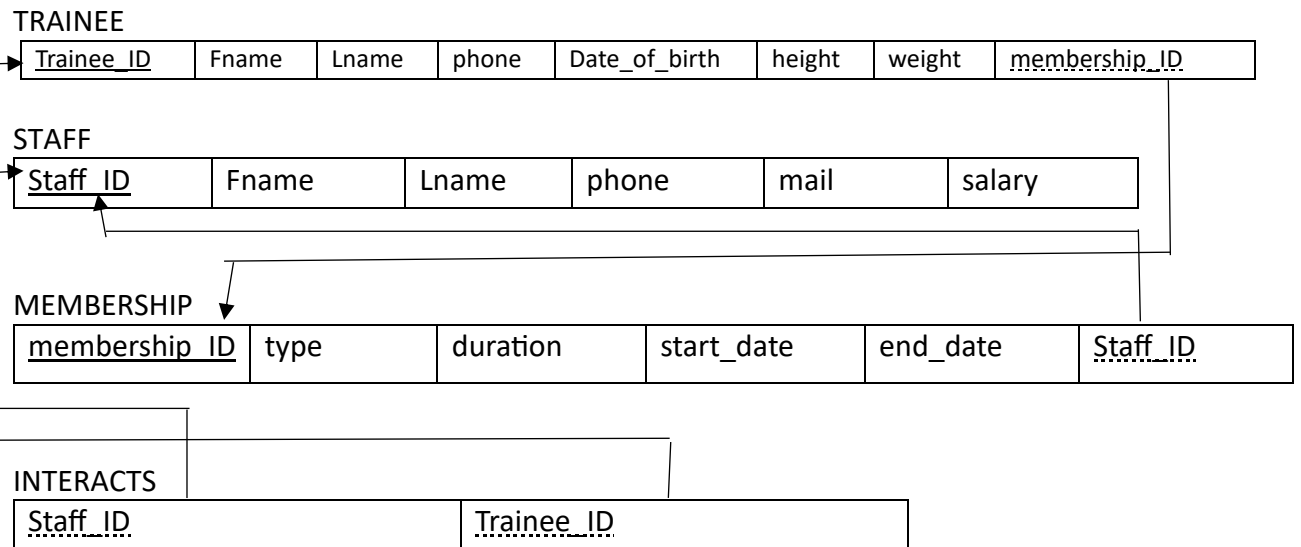


Figure1 ERD



1.5 Relational Schema diagram





Section 2



Database:

```
1 • create database GalymSEL_GYM;
2
3 • show databases;
4 • use GalymSEL_GYM;
5
6 • create table trainee (
7     trainee_ID int primary key auto_i
8     membership_ID int not null,
9     Fname varchar(60) not null,
10    Lname varchar(60) not null,
11    phone varchar(10) UNIQUE,
12    date of birth datetime not null.
```

Result Grid | Filter Rows: | Export

	Database
▶	galymsel_gym

Figure2 database



Trainee Table:

Galym SEL GYM SQL* Administration - Server Status

Limit to 1000 rows

```
6 • create table trainee (  
7     trainee_ID int primary key auto_increment,  
8     membership_ID int not null,  
9     Fname varchar(60) not null,  
10    Lname varchar(60) not null,  
11    phone varchar(10) UNIQUE,  
12    date_of_birth datetime not null,  
13    height double CHECK (height > 0),  
14    weight double CHECK (weight > 0),  
15    foreign key (membership_ID) references  
16    membership(membership_ID)  
17 );  
18 • describe trainee;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
►	trainee_ID	int	NO	PRI	NULL	auto_increment
	membership_ID	int	NO	MUL	NULL	
	Fname	varchar(60)	NO		NULL	
	Lname	varchar(60)	NO		NULL	
	phone	varchar(10)	YES	UNI	NULL	
	date_of_birth	datetime	NO		NULL	
	height	double	YES		NULL	
	weight	double	YES		NULL	

Figure 3 trainee table



Trainee Table Value:

```
77 • INSERT INTO trainee (membership_ID, Fname, Lname, phone, date_of_birth, height, weight) VALUES
78 (1, 'Safaa', 'Sobh', '0508815213', '2006-07-10', 166, 72.3),
79 (2, 'Sahar', 'Hawsawi', '0559876543', '1992-07-25', 162.0, 58.5),
80 (3, 'Elham', 'Alzahrani', '0561112233', '1988-11-02', 188.2, 90.1),
81 (4, 'Layan', 'Alshreef', '0544455667', '2001-01-15', 168.4, 64.0),
82 (5, 'Ruba', 'Al-Jamal', '0533322110', '1998-05-30', 180.0, 81.7);
83
84 • select * from trainee;
```

trainee_ID	membership_ID	Fname	Lname	phone	date_of_birth	height	weight
5010	1	Safaa	Sobh	0508815213	2006-07-10 00:00:00	166	72.3
5011	2	Sahar	Hawsawi	0559876543	1992-07-25 00:00:00	162	58.5
5012	3	Elham	Alzahrani	0561112233	1988-11-02 00:00:00	188.2	90.1
5013	4	Layan	Alshreef	0544455667	2001-01-15 00:00:00	168.4	64
5014	5	Ruba	Al-Jamal	0533322110	1998-05-30 00:00:00	180	81.7
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Figure 4 trainee table values



Staff Table:

The screenshot shows a database management tool interface. The top toolbar includes icons for file operations, execution, search, and window management, along with a 'Limit to 1000 rows' dropdown. The main area displays SQL code for creating and managing the 'Staff' table. Below the code, the 'Result Grid' tab is active, showing a table with 7 columns: Field, Type, Null, Key, Default, and Extra. The table lists the columns of the 'Staff' table: staff_ID (int, NO, PRI, NULL, auto_increment), Fname (varchar(50), NO, NULL), Lname (varchar(50), NO, NULL), phone (varchar(10), NO, NULL), mail (varchar(100), YES, UNI, NULL), and salary (double, YES, NULL).

```
26 • create table Staff (  
27     staff_ID int PRIMARY KEY auto_increment,  
28     Fname varchar(50) not null,  
29     Lname varchar(50) not null,  
30     phone varchar(10) not null,  
31     mail varchar(100) UNIQUE,  
32     salary double CHECK (salary > 0)  
33 );  
34  
35  
36 • alter table Staff auto_increment = 800;  
37 • describe Staff;  
38 • select * from Staff;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	Field	Type	Null	Key	Default	Extra
▶	staff_ID	int	NO	PRI		auto_increment
	Fname	varchar(50)	NO			
	Lname	varchar(50)	NO			
	phone	varchar(10)	NO			
	mail	varchar(100)	YES	UNI		
	salary	double	YES			

Figure 5 staff table



Staff Table Value:

```
87 • insert into Staff (Fname, Lname, phone, mail, salary) values
88   ('Laila', 'Malak', '0501111111', 'Malak@gmail.com', 12000),
89   ('Lama', 'Sama', '0552222222', 'sama@gmail.com', 13000),
90   ('Noura', 'Shumuk', '0564444444', 'Shumuk@gmail.com', 14000),
91   ('Fahda', 'Ahmed', '0543333333', 'f.ahmed@gmail.com', 12500),
92   ('Sara', 'Salem', '0595555555', 's.salem@gmail.com', 13500);
93 • describe staff;
94 • select * from staff;
95
96
```

Result Grid

Filter Rows:

Edit:

Export/Import:

W

	staff_ID	Fname	Lname	phone	mail	salary
	800	Laila	Malak	0501111111	Malak@gmail.com	12000
	801	Lama	Sama	0552222222	sama@gmail.com	13000
	802	Noura	Shumuk	0564444444	Shumuk@gmail.com	14000
	803	Fahda	Ahmed	0543333333	f.ahmed@gmail.com	12500
	804	Sara	Salem	0595555555	s.salem@gmail.com	13500
	NULL	NULL	NULL	NULL	NULL	NULL

Figure6 staff table values



Membership Table:

```
42 • create table membership (  
43     membership_ID int primary key auto_increment,  
44     staff_ID int,  
45     type enum ('trial', 'monthly', 'yearly') DEFAULT 'monthly',  
46     duration enum ('2 weeks', '4 weeks', '12 months') not null,  
47     start_date datetime,  
48     expiration_date datetime NOT NULL,  
49     foreign key (staff_ID) references  
50     Staff(staff_ID),  
51     constraint check_dates CHECK (expiration_date > start_date)  
52 );  
53  
54 • alter table membership auto_increment = 1;  
55 • describe membership;
```

Field	Type	Null	Key	Default	Extra
membership_ID	int	NO	PRI	NULL	auto_increment
staff_ID	int	YES	MUL	NULL	
type	enum('trial','monthly','yearly')	YES		monthly	
duration	enum('2 weeks','4 weeks','12 months')	NO		NULL	
start_date	datetime	YES		NULL	
expiration_date	datetime	NO		NULL	

Figure 7 membership table



Membership Table Values:

```
95
96 • insert into membership (staff_ID ,type, duration, start_date, expiration_date) values
97 (800,'trial', '2 weeks', '2025-09-25 13:30:00','2025-10-9 13:30:00'),
98 (801,'yearly', '12 months', '2025-03-12 09:30:00','2026-03-12 09:30:00'),
99 (802,'monthly', '4 weeks', '2025-01-01 15:39:33','2025-02-01 15:39:33'),
100 (803,'yearly', '12 months', '2025-12-9 10:43:00','2026-12-9 10:43:00'),
101 (804,'monthly', '4 weeks', '2026-02-08 20:50:33','2026-03-08 20:50:33');
102 • describe membership;
103 • select * from membership;
104
```

	membership_ID	staff_ID	type	duration	start_date	expiration_date
▶	1	800	trial	2 weeks	2025-09-25 13:30:00	2025-10-09 13:30:00
	2	801	yearly	12 months	2025-03-12 09:30:00	2026-03-12 09:30:00
	3	802	monthly	4 weeks	2025-01-01 15:39:33	2025-02-01 15:39:33
	4	803	yearly	12 months	2025-12-09 10:43:00	2026-12-09 10:43:00
	5	804	monthly	4 weeks	2026-02-08 20:50:33	2026-03-08 20:50:33
*	NULL	NULL	NULL	NULL	NULL	NULL

Figure8 membership table values



Interacts Table:

```
60 • ○ create table interacts(  
61     staff_ID int NOT NULL,  
62     trainee_ID int NOT NULL,  
63     foreign key (staff_ID) references  
64     Staff(staff_ID),  
65     foreign key (trainee_ID) references  
66     trainee(trainee_ID)  
67 );  
68  
69 • describe interacts;  
70
```

Result Grid | Filter Rows: | Export: | Wrap C

	Field	Type	Null	Key	Default	Extra
▶	staff_ID	int	NO	MUL	NULL	
	trainee_ID	int	NO	MUL	NULL	

Figure 9 interacts table



Interacts Table Value:

```
108
109 • INSERT INTO interacts (staff_ID, trainee_ID) VALUES
110     (800, 5000),
111     (801, 5001),
112     (802, 5002),
113     (803, 5003),
114     (804, 5004);
115
116
117 • select * from interacts;
118
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

	staff_ID	trainee_ID
▶	800	5000
	801	5001
	802	5002
	803	5003
	804	5004

Figure 10 interacts table values



Section 3



Basic SELECT:

```
124 • SELECT Fname, Lname, weight
125 FROM trainee
126 WHERE weight > 70;
```

Result Grid

	Fname	Lname	weight
▶	Safaa	Sobh	72.3
	Elham	Alzahrani	90.1
	Ruba	Al-Jamal	81.7

Figure 11 basic select for weight

```
127
128 • SELECT Fname, Lname, salary
129 FROM Staff
130 WHERE salary > 13000;
131
132
```

Result Grid

	Fname	Lname	salary
▶	Noura	Shumuk	14000
	Sara	Salem	13500

Figure 12 basic select for salary



Aggregate Functions:

```
133
134 • SELECT Staff_ID, COUNT(Trainee_ID) AS total_trainees
135 FROM INTERACTS
136 GROUP BY Staff_ID;
137
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Staff_ID	total_trainees
800	2
802	1
803	1
804	1

Figure13 aggregate function 1

```
137
138 • SELECT AVG(salary) AS avg_salary
139 FROM STAFF;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

avg_salary
13000

Figure14 aggregate function 2

```
140
141 • SELECT SUM(salary) AS total_salary
142 FROM STAFF;
143
144
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_salary
65000

Figure15 aggregate function 3



Joins between tables:

```
144
145 • SELECT t.Fname, t.Lname, m.type, m.duration, m.expiration_date
146 FROM trainee t LEFT JOIN membership m ON t.membership_ID = m.membership_ID;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	Fname	Lname	type	duration	expiration_date
▶	Safaa	Sobh	trial	2 weeks	2025-10-09 13:30:00
	Sahar	Hawsawi	yearly	12 months	2026-03-12 09:30:00
	Elham	Alzahrani	monthly	4 weeks	2025-02-01 15:39:33
	Layan	Alshreef	yearly	12 months	2026-12-09 10:43:00
	Ruba	Al-Jamal	monthly	4 weeks	2026-03-08 20:50:33

Figure16 left join

```
147
148 • SELECT s.staff_ID, s.Fname AS Staff_Fname, t.trainee_ID AS Trainee_ID, t.Fname AS Trainee_Fname, t.Lname AS Trainee_Lname
149 FROM interacts i JOIN trainee t ON i.trainee_ID = t.trainee_ID RIGHT JOIN Staff s ON i.staff_ID = s.staff_ID;
150
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: |

	staff_ID	Staff_Fname	Trainee_ID	Trainee_Fname	Trainee_Lname
▶	800	Laila	5000	Safaa	Sobh
	800	Laila	5001	Sahar	Hawsawi
	801	Lama	NULL	NULL	NULL
	802	Noura	5002	Elham	Alzahrani
	803	Fahda	5003	Layan	Alshreef
	804	Sara	5004	Ruba	Al-Jamal

Figure17 right join



Subquery, View:

```
152 • SELECT Fname, Lname, weight
153 FROM trainee
154 WHERE weight > (
155     SELECT AVG(weight)
156     FROM trainee
157 );
```

Result Grid	Filter Rows:	Export:	Wrap Cell
Fname	Lname	weight	
Elham	Alzahrani	90.1	
Ruba	Al-Jamal	81.7	

Figure18 subquery

```
158
159 • CREATE VIEW trainee_membership_view AS
160 SELECT
161     t.trainee_ID,
162     t.Fname AS trainee_first_name,
163     t.Lname AS trainee_last_name,
164     m.type AS membership_type,
165     m.duration,
166     s.Fname AS staff_first_name
167 FROM trainee t
168 JOIN membership m ON t.membership_ID = m.membership_ID
169 JOIN Staff s ON m.staff_ID = s.staff_ID;
170
171
172 • SELECT * FROM trainee_membership_view;
173
```

Result Grid			 Filter Rows:	Export: 	Wrap Cell Content: 	
	trainee_ID	trainee_first_name	trainee_last_name	membership_type	duration	staff_first_name
▶	5000	Safaa	Sobh	trial	2 weeks	Laila
	5001	Sahar	Hawsawi	yearly	12 months	Lama
	5002	Elham	Alzahrani	monthly	4 weeks	Noura
	5003	Layan	Alshreef	yearly	12 months	Fahda
	5004	Ruba	Al-Jamal	monthly	4 weeks	Sara

Figure19 view